

MODULE-02 Probability Distributions

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Random variables :-

In a random experiment, if a real variable is associated with every outcome then it is called a random variable or stochastic variable.

Random variables are usually denoted by $X, Y, Z \dots$ & it may be noted that different random variables are associated with the sample space S .

The set of all real numbers of a random variable X is called the range of X .

Ex:- while tossing a coin, suppose that the value 1 is associated for the outcome 'head' & 0 for the outcome 'tail'. We have the sample space

$S = \{H, T\}$ & if X is the random variable then

$$X(H) = 1 \quad \& \quad X(T) = 0$$

$$\text{Range of } X = \{0, 1\}$$

There are two kinds of random variable.

1. Discrete random variable.
2. Continuous random variable.

Discrete Random variable

✓ If a random variable takes finite or countably infinite number of values, then it is called a discrete random variable.

Ex:- 1) Tossing a coin & observing the outcome.

2) Throwing a 'die' & observing the number on face.

Probability Mass Function:-

probability mass function is the probability distribution of a discrete random variable & provides the possible values & their associated probabilities.

✓ If for each value x_i of a discrete random variable X , we assign a real number $p(x_i)$ such that

1. $p(x_i) \geq 0$

2. $\sum_{i \in I} p(x_i) = 1$

3. $0 \leq p(x_i) \leq 1$

4. Mean, $\mu = \sum_{i=1}^n x_i p(x_i)$

5. Variance $\sigma^2 = \sum_{i=1}^n x_i^2 p(x_i) - \mu^2$

Problems.

1. A coin is tossed twice. A random variable X represents the number of heads turning up. Find the discrete probability distribution for X . Also find its mean & variance.

Soln Given:- coin is tossed twice

$\therefore$ Sample Space $S = \{HH, HT, TH, TT\}$

Thus we have, $p(HH) = \frac{1}{4}$, $p(HT) = \frac{1}{4}$, $p(TH) = \frac{1}{4}$, $p(TT) = \frac{1}{4}$

A random variable ' X ' represents the number of heads turning up.

$\therefore X = \{0, 1, 2\}$

$P(X=0, \text{ i.e. no head}) = p(TT) = \frac{1}{4}$

$P(X=1, \text{ i.e. 1 head}) = p(HT) + p(TH) = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$

$P(X=2, \text{ i.e. 2 heads}) = p(HH) = \frac{1}{4}$

The discrete probability distribution is as follows:-

$X = x_i$	0	1	2
$P(x_i)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

1. $p(x_i) > 0$

2. $\sum p(x_i) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$

$\therefore \sum p(x_i) = 1$

3. Mean

$$\mu = \sum x_i \cdot p(x_i)$$

$$= 0\left(\frac{1}{4}\right) + 1\left(\frac{1}{2}\right) + 2\left(\frac{1}{4}\right)$$

$$\therefore \mu = 1$$

Variance,

$$\sigma^2 = \sum_{i=1}^n x_i^2 p(x_i) - \mu^2$$

$$= \left(0\left(\frac{1}{4}\right) + 1^2\left(\frac{1}{2}\right) + 2^2\left(\frac{1}{4}\right) \right) - 1^2$$

$$= \frac{1}{2} + 1 - 1$$

$$\therefore \boxed{\text{Variance} = \frac{1}{2}}$$

2. A random variable x has the f.f probability function

x	-3	-2	-1	0	1	2	3
$p(x)$	k	$2k$	$3k$	$4k$	$3k$	$2k$	k

Find the value of k & calculate the mean & standard deviation. Also find $p(x \leq 1)$, $p(x > 1)$ & $p(-1 < x \leq 2)$.

Solⁿ: The probability distribution is valid if $p(x) \geq 0$ & $\sum p(x) = 1$.

Hence we must have $k \geq 0$.

$$p(-3) + p(-2) + p(-1) + p(0) + p(1) + p(2) + p(3) = 1$$

$$k + 2k + 3k + 4k + 3k + 2k + k = 1$$

$$16k = 1$$

$$\boxed{k = 1/16}$$

The discrete probability distribution is as follows

x	-3	-2	-1	0	1	2	3
$p(x)$	$1/16$	$2/16$	$3/16$	$4/16$	$3/16$	$2/16$	$1/16$

Mean:

$$\mu = \sum x_i \cdot p(x_i)$$

$$= -3 \times \frac{1}{16} + (-2) \left(\frac{2}{16} \right) + (-1) \left(\frac{3}{16} \right) + 0 \left(\frac{4}{16} \right) + 1 \left(\frac{3}{16} \right) + 2 \left(\frac{2}{16} \right) + 3 \left(\frac{1}{16} \right)$$

$$\boxed{\mu = 0}$$

Variance

$$\sigma^2 = \sum x_i^2 p(x_i) - \mu^2$$

$$= 9 \left(\frac{1}{16} \right) + 4 \left(\frac{2}{16} \right) + 1 \left(\frac{3}{16} \right) + 0 \left(\frac{4}{16} \right) + 1 \left(\frac{3}{16} \right) + 4 \left(\frac{2}{16} \right) + 9 \left(\frac{1}{16} \right) - 0$$

$$\boxed{\therefore \sigma^2 = \frac{5}{2}}$$

$$\boxed{\text{Standard deviation } \sigma = \sqrt{\frac{5}{2}}}$$

$$\text{i). } p(x \leq 1) = p(-3) + p(-2) + p(-1) + p(0) + p(1)$$

$$= \frac{1}{16} + \frac{2}{16} + \frac{3}{16} + \frac{4}{16} + \frac{3}{16}$$

$$\boxed{p(x \leq 1) = \frac{13}{16}}$$

$$\text{ii) } p(x > 1) = p(2) + p(3)$$

$$= \frac{2}{16} + \frac{1}{16}$$

$$\boxed{p(x > 1) = \frac{3}{16}}$$

$$\text{iii) } p(-1 < x \leq 2) = p(0) + p(1) + p(2)$$

$$= \frac{4}{16} + \frac{3}{16} + \frac{2}{16}$$

$$\boxed{p(-1 < x \leq 2) = \frac{9}{16}}$$

Q3 The p.d.f of a variate x is given by the following table

x	0	1	2	3	4	5	6
$p(x)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$

Find the value of k & calculate its mean & standard deviation. Also find $p(x \geq 5)$ & $p(3 < x \leq 6)$.

Soln. The probability distribution is valid if $p(x) \geq 0$ &

$$\sum p(x) = 1$$

Hence we must have $k \geq 0$ &

$$p(0) + p(1) + p(2) + p(3) + p(4) + p(5) + p(6) = 1$$

$$k + 3k + 5k + 7k + 9k + 11k + 13k = 1$$

$$\therefore 49k = 1$$

$$k = \frac{1}{49}$$

The discrete probability distribution is as follows

x	0	1	2	3	4	5	6
$p(x)$	$\frac{1}{49}$	$\frac{3}{49}$	$\frac{5}{49}$	$\frac{7}{49}$	$\frac{9}{49}$	$\frac{11}{49}$	$\frac{13}{49}$

Mean

$$\mu = \sum x_i \cdot p(x_i)$$

$$= 0\left(\frac{1}{49}\right) + 1\left(\frac{3}{49}\right) + 2\left(\frac{5}{49}\right) + 3\left(\frac{7}{49}\right) + 4\left(\frac{9}{49}\right) + 5\left(\frac{11}{49}\right) + 6\left(\frac{13}{49}\right)$$

$$\mu = \frac{29}{7}$$

Variance. $\sigma^2 = \sum x_i^2 \cdot p(x_i) - \mu^2$

$$= \left\{ 0\left(\frac{1}{49}\right) + 1\left(\frac{3}{49}\right) + 4\left(\frac{5}{49}\right) + 9\left(\frac{7}{49}\right) + 16\left(\frac{9}{49}\right) + 25\left(\frac{11}{49}\right) + 36\left(\frac{13}{49}\right) \right\} - \frac{841}{49}$$

$$= \frac{973}{49} - \frac{841}{49} = \frac{132}{49}$$

Standard deviation

$$\sigma = \sqrt{\sigma^2}$$

$$\sigma = \sqrt{\frac{132}{49}}$$

$$\therefore \sigma = \frac{\sqrt{132}}{7}$$

$$\begin{aligned} \text{i) } P(X \geq 5) &= P(5) + P(6) \\ &= \frac{11}{49} + \frac{13}{49} \\ &= \frac{24}{49} \end{aligned}$$

$$\begin{aligned} \text{ii) } P(3 < X \leq 6) &= P(4) + P(5) + P(6) \\ &= \frac{9}{49} + \frac{11}{49} + \frac{13}{49} \\ &= \frac{33}{49} \end{aligned}$$

4. A random variable X has the following probability function.

x_i	-2	-1	0	1	2	3
$P(x_i)$	0.1	K	0.2	$2K$	0.3	K

Find the value of K & calculate the mean & variance.

soln. Probability distribution is valid if $P(x) \geq 0$ & $\sum P(x) = 1$.

Hence we must have $K \geq 0$

$$P(-2) + P(-1) + P(0) + P(1) + P(2) + P(3) = 1$$

$$0.1 + K + 0.2 + 2K + 0.3 + K = 1$$

$$0.6 + 4K = 1$$

$$4K + 0.6 = 1$$

$$4K = 1 - 0.6$$

$$K = \frac{0.4}{4}$$

$$\boxed{K = 0.1}$$

x_i	-2	-1	0	1	2	3
$p(x_i)$	0.1	0.1	0.2	0.2	0.3	0.1

Mean, $\mu = \sum x_i \cdot p(x_i)$

$$= (-2)(0.1) + (-1)(0.1) + 0(0.2) + (1)(0.2) + 2(0.3) + 3(0.1)$$

$$\mu = \frac{4}{5} = 0.8$$

Variance $\sigma^2 = \sum x_i^2 p(x_i) - \mu^2$

$$= \{4(0.1) + 1(0.1) + 0(0.2) + 1(0.2) + 4(0.3) + 9(0.1)\} - 0.64$$

$$\sigma^2 = 2.16$$

Standard deviation

$$\sigma = \sqrt{2.16}$$

$$\sigma = 1.4696$$

5. The p.d.f of a variate x is given by the following table
Find the value of K . Find the probability distribution & the distribution function of x . Also Evaluate $p(x < 6)$, $p(x \geq 6)$ & $p(3 < x \leq 6)$.

x	0	1	2	3	4	5	6	7
$p(x)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

Soln. The probability distribution is valid if $p(x) \geq 0$ & $\sum p(x) = 1$

Hence we must have $K \geq 0$ &

$$p(0) + p(1) + p(2) + p(3) + p(4) + p(5) + p(6) + p(7) = 1$$

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$10k^2 + 9k = 1$$

$$10k^2 + 9k - 1 = 0$$

$$10k^2 + 10k - 1k - 1 = 0$$

$$10k(k+1) - 1(k+1) = 0$$

$$(10k-1)(k+1) = 0$$

$$k = \frac{1}{10} \quad k = -1$$

$k = -1$, is rejected since probability cannot be negative.

$$\therefore k = \frac{1}{10}$$

The probability distribution is given by

x	0	1	2	3	4	5	6	7
$p(x)$	0	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{1}{100}$	$\frac{2}{100}$	$\frac{17}{100}$

$$\begin{aligned} \text{i) } p(x < 6) &= 1 - p(6) - p(7) \\ &= 1 - \frac{2}{100} - \frac{17}{100} \\ &= \frac{81}{100} \end{aligned}$$

$$\begin{aligned} \text{iii) } p(3 < x \leq 6) &= p(4) + p(5) + p(6) \\ &= \frac{3}{10} + \frac{1}{100} + \frac{2}{100} \\ &= \frac{33}{100} \end{aligned}$$

$$\begin{aligned} \text{ii) } p(x \geq 6) &= p(6) + p(7) \\ &= \frac{2}{100} + \frac{17}{100} \\ &= \frac{19}{100} \end{aligned}$$

Continuous probability Distribution.

Probability Density function.

probability density function is the probability distribution of a continuous random variable & provides the possible values & their associated probabilities infinitely.

If for every x belonging to the range of a continuous random variable X , we assign a real number $f(x)$ satisfying.

01. $p(x \geq 0)$ or $f(x) \geq 0$.

2. $\int_{-\infty}^{\infty} f(x) dx = 1$

3. Mean $\mu = \int_{-\infty}^{\infty} x f(x) dx$.

4. Variance $\sigma^2 = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$

5. $p(a \leq x \leq b) = p(a < x \leq b) = p(a \leq x < b)$
 $= p(a < x < b) = \int_a^b f(x) dx$.

Continuous random variable :- A random variable takes non countable infinite number of values.

Ex:- weight of articles
length of nails produced by a machine.

problems.

D. A random variable x has the following density function

$$p(x) = \begin{cases} kx^2, & -3 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

Evaluate k & find

i) $p(1 \leq x \leq 2)$

ii) $p(x \leq 2)$

iii) $p(x > 1)$

Soln $p(x) \geq 0$ if $k \geq 0$ & also we must have

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

ie

$$\int_{-3}^3 kx^2 dx = 1$$

$$k \left[\frac{x^3}{3} \right]_{-3}^3 = 1$$

$$\frac{k}{3} [27 + 27] = 1$$

$$18k = 1$$

$$k = \frac{1}{18}$$

i) $p(1 \leq x \leq 2) = \int_1^2 kx^2 dx = k \left[\frac{x^3}{3} \right]_1^2 =$

$$= \frac{1}{18 \times 3} [2^3 - 1^3]$$

$$= \frac{7}{54}$$

ii) $p(x \leq 2) = \int_{-3}^2 kx^2 dx$

$$= k \left[\frac{x^3}{3} \right]_{-3}^2$$

$$= \frac{1}{18 \times 3} [2^3 - (-3)^3]$$

$$= \frac{1}{54} [8 + 27]$$

$$P(X \leq 2) = \frac{35}{54}$$

$$\begin{aligned} \text{ii) } P(X > 1) &= \int_1^3 kx^2 dx \\ &= \left[\frac{1}{18} \cdot \frac{x^3}{3} \right]_1^3 \\ &= \frac{1}{18 \times 3} [27 - 1] \end{aligned}$$

$$P(X > 1) = \frac{13}{27}$$

2. $\Rightarrow$ If x is a continuous random variable with the probability function given by

$$f(x) = \begin{cases} kx^2 & 0 \leq x < 2 \\ 2kx & 2 \leq x < 4 \\ 6k & 4 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of k & mean.

Soln

$f(x) \geq 0$ if $k \geq 0$ and also we must have

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_0^2 kx^2 dx + \int_2^4 2kx dx + \int_4^6 6k dx = 1$$

$$k \cdot \left[\frac{x^3}{3} \right]_0^2 + 2k \left[\frac{x^2}{2} \right]_2^4 + 6k \left[x \right]_4^6 = 1$$

$$\frac{k}{3} \cdot 8 + \frac{2k}{2} (4^2 - 2^2) + 6k(6 - 4) = 1$$

$$\frac{k}{3} \cdot 8 + \frac{2k}{3} [16-4] + 6k(2) = 1$$

$$\frac{8k}{3} + 12k + 12k = 1$$

$$3 \times \frac{8k}{3} + \frac{8k}{3} = 1$$

$$\frac{72k + 8k}{3} = 1$$

$$\frac{80k}{3} = 1$$

$$\boxed{k = \frac{3}{80}}$$

Mean $\mu = \int_{-\infty}^{\infty} x f(x)$

$$= \int_0^2 x(kx^2) dx + \int_2^4 x(2kx) dx + \int_4^6 x(6k) dx$$

$$= \int_0^2 k \cdot x^3 dx + \int_2^4 2k x^2 dx + \int_4^6 6k x dx$$

$$= k \cdot \frac{x^4}{4} \Big|_0^2 + 2k \cdot \frac{x^3}{3} \Big|_2^4 + 6k \cdot \frac{x^2}{2} \Big|_4^6$$

$$= \frac{k}{4} [2^4] + \frac{2k}{3} [4^3 - 2^3] + \frac{6k}{2} [6^2 - 4^2]$$

$$= 4k + \frac{2k}{3} \cdot 56 + 3k[20]$$

$$= 4k + \frac{112k}{3} + 60k$$

$$= \frac{12k + 112k + 180k}{3}$$

$$= \frac{304k}{3}$$

$$= \frac{304 \times \cancel{3}}{80 \times \cancel{3}} = 3.8 //$$

5. In 800

- 03 Find the constant K such that $f(x) = \begin{cases} Kx^2, & 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$ is a probability density function. Also compute
- i) $p(1 < x < 2)$
 - ii) $p(x \leq 1)$
 - iii) $p(x > 1)$
 - iv) mean
 - v) variance

Binomial Distributions.

Binomial distribution gives the probability of getting exactly x successes in n independent trials, each with probability of success p .

Conditions

Binomial applies when

1. Fixed number of trials (n)
2. Each trial has only two outcomes
Success (p), Failure ($q=1-p$)
3. Probability of success is same for every trial.
4. All trials are independent.

Formula.

$$P(X=x) = {}^nC_x p^x (1-p)^{n-x}, x=0,1,2,\dots,n.$$

or

$${}^nC_x p^x q^{n-x}$$

n = number of trials

x = number of successes

p = probability of success

$q = 1-p$ = probability of failure.

Properties

- Mean = np
- Variance = $np(1-p)$ / npq
- Standard deviation $\sigma = \sqrt{npq}$

probability distribution table.

x	0	1	2
$p(x)$	$nC_0 p^0 q^{n-0}$	$nC_1 p^1 q^{n-1}$	$nC_2 p^2 q^{n-2}$

Binomial Distribution table using calci.

1. clear CALCI [Shift-A-3- $\square$]
2. press mode $\rightarrow$ 7 $\rightarrow$ Table
3. Define function

$$f(x) = nC_x p^x q^{n-x} \quad [n=4, p=0.5, q=0.5]$$

$$= 2[\text{shift} \div] \text{Alpha} \times (0.5)^{\text{Alpha} \times} \times (0.5)^{\text{Alpha} \times}$$

$$=$$

Start = 0
End = 4
Step = 1

x	0	1	2	3	4
$p(x)$	0.0625	0.25	0.375	0.25	0.0625

Problems-

1. When a coin is tossed four times, find the probability^{ny} of getting
- Exactly one head
 - Atmost three heads
 - Atleast two heads.

Soln

Number of trials: $n = 4$

Probability of success (Head): $p = \frac{1}{2}$

Probability of failure (Tail) = $q = 1 - p = \frac{1}{2}$.

Formula:-

$$p(x) = {}^nC_x p^x q^{n-x} = {}^4C_x (0.5)^x (0.5)^{4-x}$$

1. $p(0) = {}^4C_0 (0.5)^0 (0.5)^{4-0} = 0.0625$

2. $p(1) = {}^4C_1 (0.5)^1 (0.5)^{4-1} = 0.25$

3. $p(2) = {}^4C_2 (0.5)^2 (0.5)^{4-2} = 0.375$

4. $p(3) = {}^4C_3 (0.5)^3 (0.5)^{4-3} = 0.25$

5. $p(4) = {}^4C_4 (0.5)^4 (0.5)^{4-4} = 0.0625$

probability distribution table

a) Exactly 1 head

$$p(x=1) = 0.25$$

x	0	1	2	3	4
$p(x)$	0.0625	0.25	0.375	0.25	0.0625

b) Atmost three heads

$$\begin{aligned} p(x \leq 3) &= p(0) + p(1) + p(2) + p(3) \\ &= 0.0625 + 0.25 + 0.375 + 0.25 \\ &= 0.9375 \end{aligned}$$

c) Atleast two heads

$$\begin{aligned} p(x \geq 2) &= p(2) + p(3) + p(4) \\ &= 0.375 + 0.25 + 0.0625 \\ &= 0.6875 \end{aligned}$$

2. The probability that a pen manufactured by a factory be defective is $\frac{1}{10}$. If 12 such pens are manufactured what is the probability that

- i) Exactly 2 defective.
- ii) At least 2 are defective
- iii) None of them are defective.

Solⁿ

$$n = 12$$

$$p = \frac{1}{10} = 0.1$$

$$q = 1 - p = 1 - 0.1 = 0.9$$

$$\text{Formula: } p(x) = {}^n C_x p^x q^{n-x} = {}^{12} C_x (0.1)^x (0.9)^{12-x}$$

Probability Distribution table

- i) Exactly 2 defective.

$$\begin{aligned} p(x=2) &= {}^{12} C_2 (0.1)^2 (0.9)^{12-2} \\ &= 0.2301 \end{aligned}$$

- ii) At least 2 are defective

$$\begin{aligned} p(x \geq 2) &= 1 - p(x < 2) \\ &= 1 - p(0) - p(1) \\ &= 1 - 0.2824 = 0.7176 \\ &= 0.7176 \end{aligned}$$

- iii) None of them are defective

$$\begin{aligned} p(x=0) &= {}^{12} C_0 (0.1)^0 (0.9)^{12-0} \\ &= 0.2824 \end{aligned}$$

- 3 The probability that a person aged 60 years will live up to 70 is 0.65. What is the probability that out of 10 persons aged 60, at least 7 of them will live up to 70.

$X \rightarrow$ no. of persons aged 60 years living upto 70 years.

$$p = 0.65$$

$$q = 1 - p = 0.35$$

$$n = 10.$$

$$\text{Formula: } p(x) = {}^n C_x p^x q^{n-x} \\ = {}^{10} C_x (0.65)^x (0.35)^{10-x}$$

$$\begin{aligned} \text{i) } p(\text{atleast } 7) &= p(x \geq 7) \\ &= p(7) + p(8) + p(9) + p(10) \\ &= {}^{10} C_7 (0.65)^7 (0.35)^3 + {}^{10} C_8 (0.65)^8 (0.35)^2 + \\ &\quad {}^{10} C_9 (0.65)^9 (0.35)^1 + {}^{10} C_{10} (0.65)^{10} (0.35)^0 \\ &= 0.2522 + 0.1756 + 0.0724 + 0.0134 \\ &= \underline{\underline{0.5136}} \end{aligned}$$

4. The number of telephone lines busy at an instant of time is binomial variate with probability 0.1 that a line is busy. If 10 lines are chosen at random, what is the probability that

- i) no line is busy
- ii) All lines are busy
- iii) Atleast one line is busy
- iv) Atmost 2 lines are busy

Soln

$$n = 10$$

$$p = 0.1 \text{ (line is busy)}$$

$$q = 0.9 \text{ (not busy)}$$

$$X = \text{number of lines busy.}$$

$$\text{Formula: } {}^nC_x p^x q^{n-x} = {}^{10}C_x (0.1)^x (0.9)^{10-x}.$$

i) No line is busy

$$P(X=0) = {}^{10}C_0 (0.1)^0 (0.9)^{10-0} = 0.3487.$$

ii) All lines are busy

$$P(X=10) = {}^{10}C_{10} (0.1)^{10} (0.9)^{10-10} = 1 \times 10^{-10}.$$

iii) Atleast one line busy.

$$\begin{aligned} P(X \geq 1) &= 1 - P(0) \\ &= 1 - {}^{10}C_0 (0.1)^0 (0.9)^{10-0} \\ &= 1 - 0.3486 \\ &= \end{aligned}$$

iv) Atmost 2 lines are busy

$$\begin{aligned} P(X \leq 2) &= P(0) + P(1) + P(2) \\ &= {}^{10}C_0 (0.1)^0 (0.9)^{10-0} + {}^{10}C_1 (0.1)^1 (0.9)^{10-1} + {}^{10}C_2 (0.1)^2 (0.9)^{10-2} \\ &= 0.3486 + 0.3874 + 0.1937 = 0.9297. \end{aligned}$$

Q. In 800 families with 5 children each how many families would be expected to have

i) 3 boys

ii) 5 girls

iii) Either 2 or 3 boys

iv) Atmost 2 girls by assuming probability for boys & girls.

Soln

$$n = 5$$

$$P = \text{probability of having boy} = \frac{1}{2}$$

$$Q = \text{probability of having girl} = \frac{1}{2}$$

$X \Rightarrow$ number of boys in a family

Formula

$$P(X) = {}^nC_x p^x q^{n-x} = {}^5C_x (0.5)^x (0.5)^{5-x}$$

Distribution table

x	0	1	2	3	4	5
$P(x)$	0.0312	0.1562	0.3125	0.3125	0.1562	0.0312

Total families = 800

Expected families = $800 \times P(\text{event})$

i) 3 boys

$$P(X=3) = 0.3125$$

$$\text{Expected families} = 800 \times 0.3125 \\ = 250$$

ii) 5 girls.

5 girls means 0 boys

$$P(X=0) = 0.0312$$

$$\text{Expected families} = 25$$

iii) Either 2 or 3 boys

$$\begin{aligned}P(X=2 \text{ or } 3) &= P(2) + P(3) \\&= 0.3125 + 0.3125 \\&= 0.625\end{aligned}$$

$$\text{Expected families} = 800 \times 0.625 = \underline{\underline{500}}$$

iv). Atmost 2 girls

means girls ≤ 2 with 5 children
boys ≥ 3 .

$$\begin{aligned}P(\text{boys} \geq 3) &= P(3) + P(4) + P(5) \\&= 0.3125 + 0.1562 + 0.0312 \\&= 0.4999\end{aligned}$$

$$\begin{aligned}\text{Expected families} &= 800 \times 0.4999 \\&= \underline{\underline{400}}\end{aligned}$$

6. The number of telephone

Five dice were thrown 96 times & the number of times an odd number actually turned out in the experiment is given. Fit a binomial distribution to this data & calculate Expected frequencies

$$n=5$$

$$p = \text{odd face} = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

$$q = \frac{1}{2}$$

No. of dice showing
1 or 3 or 5

observed frequency

0	1	2	3	4	5
1	10	24	35	18	8

$$n=5$$

$$p = \text{odd face} = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

$$q = \frac{1}{2}$$

Formula:- $nC_x p^x q^{n-x}$

$$P(x) = {}^5C_x (0.5)^x (0.5)^{5-x}$$

Distribution table.

x	0	1	2	3	4	5
$f(x)$	0.0312	0.1562	0.3125	0.3125	0.1562	0.0312

Since 5 dice were thrown 96 times

$$\therefore F(x) = 96 \cdot p(x)$$

When $x = 0, 1, 2, 3, 4, 5$

$$F(0) = 96 \cdot p(0) = 2.99 = 3$$

$$F(1) = 96 \cdot p(1) = 15$$

$$F(2) = 96 \cdot p(2) = 30$$

$$F(3) = 96 \cdot p(3) = 30$$

$$F(4) = 96 \cdot p(4) = 15$$

$$F(5) = 96 \cdot p(5) = 3$$

$\therefore$ The Expected theoretical frequencies are
3, 15, 30, 30, 15, 3.

Poisson distribution.

Poisson distribution is obtained as the limiting form of the binomial distribution when n is very large ($n \rightarrow \infty$) & p is very small ($p \rightarrow 0$), such that $np = m$ (a finite constant).

Conditions for Poisson distribution

1. The number of trials is very large.
2. The probability of success in each trial is very small.
3. Events occur independently
4. The mean number of successes (m) in the interval is finite.

Formula:-

$$P(X=x) = \frac{e^{-m} \cdot m^x}{x!}, \quad x = 0, 1, 2, \dots$$

where

x = number of (occurrences) successes

m = mean number of occurrences in the given interval

$e = 2.718$ (Euler's constant)

properties:

- Mean = $m = np$
- Variance = np
- Standard deviation = $\sqrt{np}$

probability distribution table.

x	0	1	2
$p(x)$	e^{-m}	$m e^{-m}$	$\frac{m^2 e^{-m}}{2!}$

problems:-

1. In a certain factory producing razor blades, the probability of any blade being defective is $\frac{1}{500}$. If blades are supplied in packets of 10, using poisson distribution determine the approximate number of packets containing
- No defective
 - 1 defective
 - Two defective blades, respectively in a consignment of 10000 packets

Soln

- probability of defective blade $p = \frac{1}{500} = 0.002$.
- Blades per packet $= 10$
- Number of packets = 10,000

The mean number of defective blades is

$$m = np = 10 \times 0.002$$

$$m = 0.02$$

poisson probabilities for a single packet

Formula:-

$$p(x) = \frac{m^x e^{-m}}{x!}$$

$$p(0) = \frac{m^0 e^{-0.02}}{0!} = 0.98019$$

$$p(1) = \frac{m^1 e^{-0.02}}{1!} = \frac{(0.02)^1 e^{-0.02}}{1!} = 0.019603$$

$$p(2) = \frac{m^2 e^{-0.02}}{2!} = \frac{(0.02)^2 e^{-0.02}}{2!} = 0.00019$$

Expected number of packets = Total packets $\times$ probability,
 $E(x) = 10,000 \times p(x)$.

i) No defective

$$P(0) = 0.98019$$

$$\text{Expected number of packets} = 10,000 \times 0.98019 = \underline{\underline{9802}}$$

ii) one defective.

$$P(1) = 0.019603$$

$$\text{Expected number of packets} = 10,000 \times 0.019603 = 196$$

iii) Two defective

$$P(2) = 0.00019$$

$$\text{Expected number of packets} = 10,000 \times 0.00019 = \underline{\underline{2}}$$

2. 2% of the fuses manufactured by a firm are found to be defective. Find the probability that a box containing 200 fuses contains

i) No defective fuses

ii) 3 or more defective fuses.

Soln $p = \frac{2}{100} = 0.02$

$$n = 200$$

$$\mu = np = 200 \times 0.02$$

$$\boxed{\mu = 4}$$

Poisson probabilities

$$\text{Formula } P(x) = \frac{m^x e^{-m}}{x!} = \frac{4^x e^{-4}}{x!}$$

$$P(0) = \frac{4^0 e^{-4}}{0!} = 0.0183$$

$$P(1) = \frac{4^1 e^{-4}}{1!} = 0.0732$$

$$P(2) = \frac{4^2 \cdot e^{-4}}{2!} = 0.1465$$

$$P(3) = \frac{4^3 \cdot e^{-4}}{3!} = 0.1953$$

$$\begin{aligned} \text{i) } P(X=0) &= P(0) \\ &= 0.0183 \end{aligned}$$

ii) 3 or more defective fuses

$$\begin{aligned} P(X \geq 3) &= 1 - P(X < 3) \\ &= 1 - P(0) - P(1) - P(2) \\ &= 1 - 0.0183 - 0.0732 - 0.1465 \\ &= 0.7622 \end{aligned}$$

3. If the probability of a bad reaction from a certain injection is 0.001, determine the chance that out of 2000 individuals, more than two will get a bad reaction

$$p = 0.001$$

$$n = 2000$$

$$\mu = np = 2000 \times 0.001$$

$$\boxed{\mu = 2}$$

Poisson probabilities

$$\text{Formula: } P(X) = \frac{\mu^x \cdot e^{-\mu}}{x!} = \frac{2^x \cdot e^{-2}}{x!}$$

$$P(0) = \frac{2^0 \cdot e^{-2}}{0!} = 0.1353$$

$$P(1) = \frac{2^1 \cdot e^{-2}}{1!} = 0.2706$$

$$P(2) = \frac{2^2 \cdot e^{-2}}{2!} = 0.2706$$

$$P(X > 2) = 1 - P(X \leq 2)$$

$$= 1 - P(0) - P(1) - P(2)$$

$$= 1 - 0.1353 - 0.2706 - 0.2706$$

$$= 0.3235$$

4. The probability that a news reader commits no mistake in reading the news is $\frac{1}{e^3}$. Find the probability that on a particular news broadcast, he commits

- only two mistakes
- more than 3 mistakes
- Atmost 3 mistakes

$$P = \frac{1}{e^3}$$

$x \rightarrow$ denotes committing a mistake Formula: $p(x) = \frac{m^x e^{-m}}{x!}$

$$p(0) = e^{-3}$$

$$\frac{m^0 e^{-m}}{0!} = e^{-3}$$

$$e^{-3} = e^{-3}$$

$$\boxed{m=3}$$

$$p(x) = \frac{3^x \cdot e^{-3}}{x!}$$

Poisson probability

$$p(0) = \frac{3^0 \cdot e^{-3}}{0!} = 0.0497$$

$$p(1) = \frac{3^1 \cdot e^{-3}}{1!} = 0.1493$$

$$p(2) = \frac{3^2 \cdot e^{-3}}{2!} = 0.224$$

$$p(3) = \frac{3^3 \cdot e^{-3}}{3!} = 0.224$$

$$i) p(x=2) = p(2) = 0.224 //$$

$$ii) p(x=1) = p(1) = 0.1493$$

$$\begin{aligned} ii) p(x \geq 3) &= 1 - p(x \leq 3) \\ &= 1 - p(0) - p(1) - p(2) - p(3) \\ &= 1 - 0.0497 - 0.1493 - 0.224 - 0.224 \\ &= 0.353 // \end{aligned}$$

$$\begin{aligned} ii) p(x \leq 3) &= p(0) + p(1) + p(2) + p(3) \\ &= 0.0497 + 0.1493 + 0.224 + 0.224 \\ &= 0.647 // \end{aligned}$$

5. The number of accidents in a year to taxi drivers in a city follows a poisson distribution with mean 3 out of 1000 taxi drivers find approximately the number of drivers with
- no accident in a year
 - More than 3 accidents in a year.

Soln:

Mean $\mu = 3$.

The poisson distribution is given by $p(x) = \frac{m^x \cdot e^{-m}}{x!} = \frac{3^x \cdot e^{-3}}{x!}$

Expected no of drivers $= 1000 \times p(x)$
 $f(x) = 1000 \times p(x)$

i) No accident

$$P(X=0) = \frac{3^0 \cdot e^{-3}}{0!} = 0.0497$$

$$f(x) = 1000 \times p(0) = 49.7 \approx 50$$

ii). $p(\text{More than 3 accidents in a year})$

$$p(x > 3) = 1 - p(x \leq 3)$$

$$= 1 - p(0) - p(1) - p(2) - p(3)$$

$$= 1 - \frac{3^0 \cdot e^{-3}}{0!} - \frac{3^1 \cdot e^{-3}}{1!} - \frac{3^2 \cdot e^{-3}}{2!} - \frac{3^3 \cdot e^{-3}}{3!}$$

$$= 1 - 0.0497 - 0.1493 - 0.2240 - 0.2240$$

$$= 0.353$$

$$f(x) = 1000 \times 0.353$$

$$= \underline{\underline{353}}$$

Normal Distribution

- Normal distribution is a continuous probability distribution.
- It can indeed be derived from the binomial distribution as a limiting case.
 - When the number of trials $n \rightarrow \infty$
 - Probability of Success p is not too close to 0 or 1 (best when $p \approx 1/2$).
 - Then the binomial distribution tends to the normal distribution.

General Equation of Normal Distribution

The probability density function of a normal distribution is:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-\frac{z^2}{2}} dz \quad -\infty < z < \infty$$

- μ = mean
- σ = standard deviation

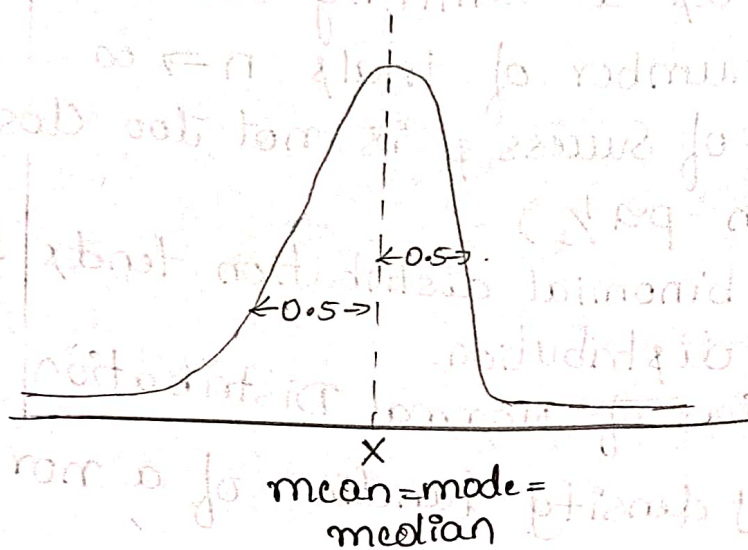
properties

1. The curve is bell-shaped & symmetrical about the mean.
2. Total area under the curve = 1
 - left side of mean = 0.5
 - Right side of mean = 0.5
3. Mean = Median = Mode (all coincide at the centre)
4. For probability calculation, first convert x into standard normal variate (z).

Standard Normal variate

$$Z = \frac{x - \mu}{\sigma}$$

- x = normal variate
- μ = mean
- σ = Standard deviation



- 1. The curve is bell-shaped & symmetrical about the mean.
- 2. Total area under the curve = 1
- 3. Left side of mean = 0.5
- 4. Right side of mean = 0.5
- 5. Mean = Median = Mode (all coincide at the center)
- 6. First moment about the mean is zero
- 7. Second moment about the mean is 1
- 8. Third moment about the mean is 0
- 9. Fourth moment about the mean is 3
- 10. Fifth moment about the mean is 0
- 11. Sixth moment about the mean is 15
- 12. Seventh moment about the mean is 0
- 13. Eighth moment about the mean is 105
- 14. Ninth moment about the mean is 0
- 15. Tenth moment about the mean is 945
- 16. Eleventh moment about the mean is 0
- 17. Twelfth moment about the mean is 10395
- 18. Thirteenth moment about the mean is 0
- 19. Fourteenth moment about the mean is 135135
- 20. Fifteenth moment about the mean is 0
- 21. Sixteenth moment about the mean is 2027025
- 22. Seventeenth moment about the mean is 0
- 23. Eighteenth moment about the mean is 30535575
- 24. Nineteenth moment about the mean is 0
- 25. Twentieth moment about the mean is 452745375

1. If X is normally distributed with mean 12 & standard deviation 4. Find the following
 [given $\phi(z) = 0.4772$]

i) $P(X \geq 20)$

ii) $P(X \leq 20)$

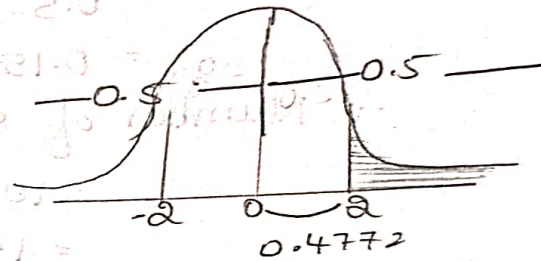
Soln

$\mu = 12 \quad \sigma = 4$

i) $P(X \geq 20)$

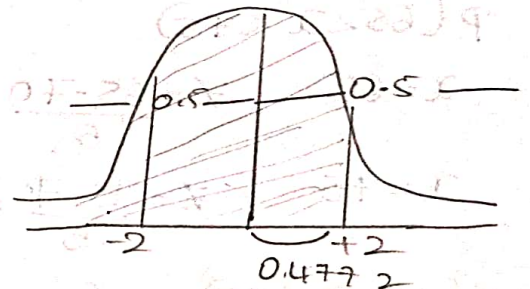
$Z = \frac{x - \mu}{\sigma} = \frac{20 - 12}{4} = \frac{08}{4} = 2$

$P(Z \geq 2) = 0.5 - \phi(2)$
 $= 0.5 - 0.4772$
 $= 0.0228$



ii) $P(X \leq 20) = P(Z \leq 2)$

$= 0.5 + \phi(2)$
 $= 0.5 + 0.4772$
 $= 0.9772$



2. The marks of 1000 students in an exam follows normal distribution with mean 70 & SD 5. Find the students whose marks will be

i) less than 65

ii) Between 65 & 75

iii) More than 75

[$\phi(z) = 0.3413$]

Soln: let x represents the marks of the students

$\mu = 70$

$SD = 5$

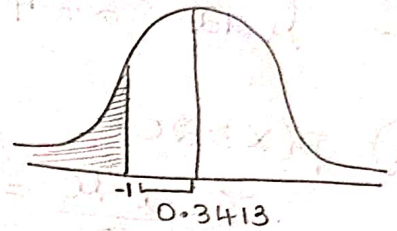
standard normal variate, $z = \frac{x - \mu}{\sigma} = \frac{x - 70}{5}$

i) less than 65

$$p(x < 65)$$

$$x = 65 \quad z = \frac{65 - 70}{5} = -1$$

$$\begin{aligned} p(x < 65) &= p(z < -1) \\ &= 0.5 - \phi(1) \\ &= 0.5 - 0.3413 \\ &= 0.1587 \end{aligned}$$



$\therefore$ Number of students scoring less than 65 marks
 $= 1000 \times 0.1587$
 $= 158.7 \approx 159.$

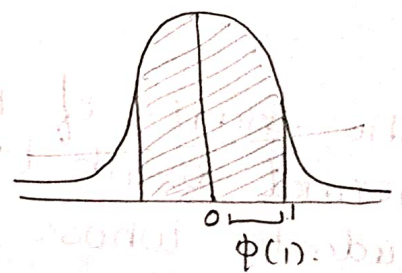
ii) Between 65 & 75

$$p(65 < x < 75)$$

$$x = 65 \quad z = \frac{65 - 70}{5} = -1$$

$$x = 75 \quad z = \frac{75 - 70}{5} = 1$$

$$\begin{aligned} p(65 < x < 75) &= p(-1 < z < 1) \\ &= 2\phi(1) \\ &= 2 \times 0.3413 \\ &= 0.6826 \end{aligned}$$



$\therefore$ Number of students scoring b/w 65 & 75
 $= 1000 \times 0.6826 = 682.6 \approx 683$ students.

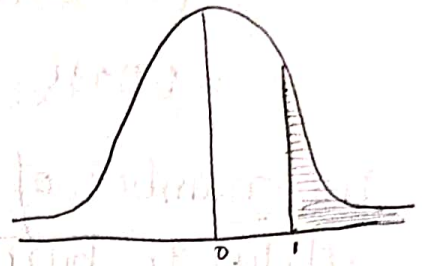
iii)

More than 75

$$P(X > 75)$$

$$x = 75 \quad z = \frac{75 - 70}{5} = 1$$

$$\begin{aligned} P(X > 75) &= P(Z > 1) \\ &= 0.5 - \phi(1) \\ &= 0.1587 \end{aligned}$$



$\therefore$ Number of students scoring more than 75 is
 $1000 \times 0.1587 = 158.7 \approx 159$

3. In a test on 2000 electric bulbs, it was found that the life of a particular make was normally distributed with an average life 2040 hours & SD of 60 hours estimate the number of bulbs likely to burn for

i) More than 2150 hours

ii) Less than 1950 hours

iii) In between 1980 & 2160 hours.

[Given $\phi(1.83) = 0.4664$, $\phi(2) = 0.4772$, $\phi(1.5) = 0.4332$]

Solⁿ: Let x represent the number of bulbs likely to burn

$$\mu = 2040$$

$$\sigma = 60$$

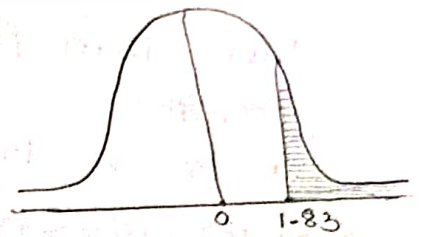
$$\text{Standard normal variate } z = \frac{x - \mu}{\sigma} = \frac{x - 2040}{60}$$

i) More than 2150 hours

$$x = 2150 \quad z = \frac{2150 - 2040}{60} = \frac{110}{60} = 1.833$$

$$P(X > 2150) = P(Z > 1.833)$$

$$\begin{aligned}
 P(Z > 1.83) &= 0.5 - \phi(1.83) \\
 &= 0.5 - 0.4664 \\
 &= 0.0336
 \end{aligned}$$



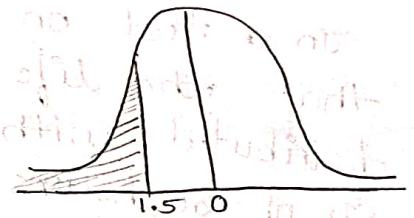
∴ The number of bulbs that are likely to burn for more than 2150 hours is $2000 \times 0.0336 = 67.2 \approx 67$ bulbs.

ii) less than 1950 hours

$$P(X < 1950)$$

$$X = 1950 \quad Z = \frac{1950 - 2040}{60} = -1.5$$

$$\begin{aligned}
 \therefore P(Z < -1.5) &= P(Z \geq 1.5) \\
 &= 0.5 - \phi(1.5) \\
 &= 0.5 - 0.4332 \\
 &= 0.0668
 \end{aligned}$$



∴ The number of bulbs that are likely to burn for less than 1950 hours

$$2000 \times 0.0668 = 133.6 \approx 134 \text{ bulbs}$$

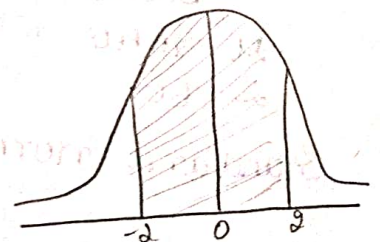
iii) b/w 1920 & 2160 hours

$$P(1920 < X < 2160)$$

$$X = 1920 \quad Z = \frac{1920 - 2040}{60} = -2$$

$$X = 2160 \quad Z = \frac{2160 - 2040}{60} = 2$$

$$\begin{aligned}
 P(-2 < Z < 2) &= 2\phi(2) \\
 &= 2 \times 0.4772 \\
 &= 0.9544
 \end{aligned}$$



∴ The number of bulbs that are likely to burn b/w 1920 & 2160 hours = $2000 \times 0.9544 = 1908.8 \approx 1909$ bulbs.